

Frank Shan

✉ g3shan@uwaterloo.ca ☎ +1-263-881-6666 in frank-shan 📄 knarfthekant 🌐 portfolio

HIGHLIGHTS

- Backend-leaning builder experienced in **data modeling**, correctness, and systems that balance speed, maintainability, and reliable data integrity in startup and research environments.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* *Sep 2024 - Apr 2029*
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* *Sep 2025 - Apr 2026*
Ziyutec Inc.

- Architected a Role-Based Access Control framework (NestJS, PostgreSQL) to support extensible user roles; balanced database **normalization** against **query performance** via denormalization and data **integrity** checks.

PROJECTS

Goodline Inventory | *Python, Next.js, PostgreSQL* [github](#) 

- Built a **full-stack inventory management system** for a local business, featuring **full-text search**, real-time stock tracking, order management, and user authentication.

SKILLS

Programming Languages: Python, TypeScript, SQL, Java, C++

AI and ML: LangGraph, RAG evaluation, DeepEval, hybrid retrieval, probabilistic modeling

Frameworks: Jinja2, prompt_toolkit, NestJS, React

Platforms: AWS, PostgreSQL, LaTeX, Linux

PROFESSIONAL CERTIFICATES

Stanford Online *December 2025*
XCS224N Natural Language Understanding [credential](#) 